Renishaw Raman Spectrometer

- Make: Renishaw
- Model: InVia
- Serial Number: H41751
- Date of manufacture: April 2006
- Current service engineer contact: Luis Cepeda
- Laser wavelength: 514 nm

Vender supplied Training pdfs

all pdfs zipped together

Standard Operating Procedure (SOP)

last updated 7/29/2021 : SOP

Service History

- 2019 laser replaced
- 7/13/2021 Dichroic mirror assembly replaced and system aligned

Laser Power at amps/watts

-

Instrument Notes

- use 3 knobs on mirror inside lower left of instrument to get laser in crosshairs in the video image, as of 7/29/21 center of crosshairs are incorrect (about 430 counts when laser at top of bottom center crosshair, static scan default settings)
- Laser's readings on standby: current: 3.3 A power = 0 mW
- Laser's readings on run (100 mW range): current = 7.2 A power = 11.1 mW
- microscope: all levers on upper right hand side are in the maximum up position
- static measurement: grating fixed, smaller spectral width but fast acquistion
- extended measurement: grating rotates to increase spectral width but this slows data acquisition, must have a minimum exposure time of 10 seconds

Renishaw Raman Microscope SOP

1) Turn on laser
    a. Switch on power supply to the laser. This is located underneath the table.
    b. Make sure switch on box is on standby
    c. Turn key to "I" and wait for laser to come on, it will take about 55 seconds
    d. Once laser is on the then switch from standby to run. Current should be around 7.2 A and power around 11.1 mW.
2) Turn on spectrometer: do this at least 20 minutes before you're planned measurement
3) Launch Wire 2.0 software (accept the default settings for referencing of the motors when the dialog box pops up by clicking ok)
4) Calibrate your wavelength by doing a quick calibration using the silicon standard
    a. Place silicon card on sample start and focus on the silicon. When focusing have the dials on the microscope in the 4,1 positions.
    b. The focal point for the 20X objective is around 3 mm from the sample surface. **Do not** approach closer than this or you'll crash the objective into the sample.
    c. After focusing switch the dials on the front of the microscope to the 1,4 positions. Check to see if the laser top of the bottom crosshair in the software (as of 7/29/2021 the center of the crosshairs was not correct). If the laser is not in the correct position notify the SIL staff and they will realign the laser.
    d. After focusing go Tools → Calibration → Quick Calibration
        i. When this is finished the dialog box should say "Quick Calibration Complete".
5) Analyze your sample
    a. Place your sample on the microscope stage
    b. If your sample is very thick you may need to move the stage down. Just below the sample stage on the right-hand side there is a lever. Before unlocking the lever move the microscope stage take all the way down using the focus knobs. Once all the way down unlock the lever and move the stage with your sample on it so that your sample is around 10 mm from the 20X objective lens and then lock the lever. If you're having difficulty focusing on your sample, mark a portion of your sample with a sharpie and focus on the sharpie. Then translate the stage to a sample point after focusing.
6) Record a spectrum
    a. Using the drop-down box on the New Measurement icon select Spectral acquisition
    b. You can collect data in two modes
        i. Static: fix spectral width and fixed monochrometer. You can specify where you want the center of your spectral width, along with a exposure time (1 second default) and signal averaging (1 spectrum by default)
        ii. Extended: you can specify the spectral width and collect a larger spectral width by rotating the monochrometer. This is a slower scan. You can also supply an exposure time (10 seconds by default) and a how many

spectrum scans you want to average together to get one spectrum (1 scan by default).

iii. Click apply and ok to accept any changes you've made
iv. Click the run icon to collect the data
v. Once finished save your data

c. If you want to modify your settings click the Setup Measurement button (has triangle on it) to modify them and then apply and ok to accept modifications. Click the run button to recollect the data.
d. If you want to abort a run in progress, click the abort button (has red beaker on it)

7) Shutting down the instrument
   a. Remove the sample and clean-up any mess you've made around the microscope
   b. Place the laser in standby
   c. Turn off the power to the laser on the remote
   d. Turn off the power to the laser power supply
   e. Turn off the power to the spectrometer
   f. Make you turn off the microscope illumination light

Renishaw plc
Spectroscopy Products Division
New Mills, Wotton-under-Edge,
Gloucestershire GL12 8JR
United Kingdom

Tel +44 (0) 1453 524524
Email raman@renishaw.com

www.renishaw.com

TM001 - Introduction to Raman spectroscopy WiRE™ 5

What is Raman scattering?

Raman scattering is named after the Indian scientist C.V. Raman who discovered the effect in 1928. If light of a single colour (wavelength) is shone on a material, most scatters off with no change in the colour of the light (Rayleigh scattered light). However a tiny fraction of the light (normally about 1 part in 10 million) is scattered with a slightly different colour (Raman scattered). This light changes colour because it exchanges energy with vibrations in the material. This makes Raman scattering an excellent tool for probing vibrations in materials.

The aim of Raman spectroscopy is to analyse the Raman scattered light and infer from it as much as possible about the chemistry and structure of the material.

More on Raman scattering

Scattering occurs when an electromagnetic wave encounters a molecule, or passes through a lattice. When light encounters a molecule, the vast majority of photons (>99.999%) are elastically scattered; this Rayleigh scattering has the same wavelength as the incident light. However, a small proportion (<0.001%) will undergo inelastic (or Raman) scattering where the scattered light undergoes a shift in energy; this shift is characteristic of the species present in the sample. This process is shown in in Figure 1.

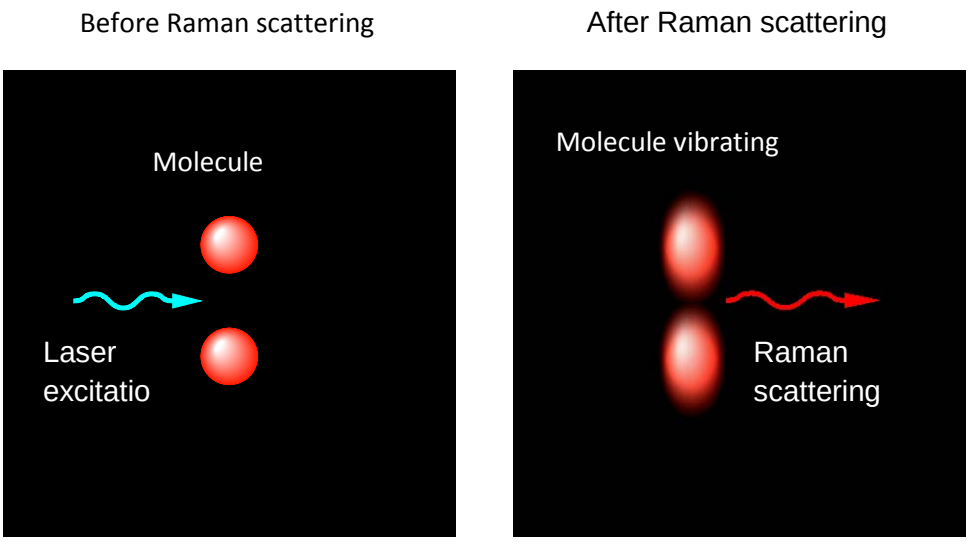


Fig. 1. Schematic diagram of the Raman effect

Figure 2 illustrates the transitions accompanying Rayleigh and Raman scattering. The electric field of the incident light distorts the molecule's electron cloud, causing it to undergo electronic transitions to a higher energy 'virtual state'; not a true quantum mechanical state of the molecule. Raman scattering results in the release of a scattered photon with different energy to the incident photon; the difference in energy is equal to the vibrational transition, ΔE. The relative intensity of stoeks ans anti-Stokes lines at room temperature is shown in Figure 3.

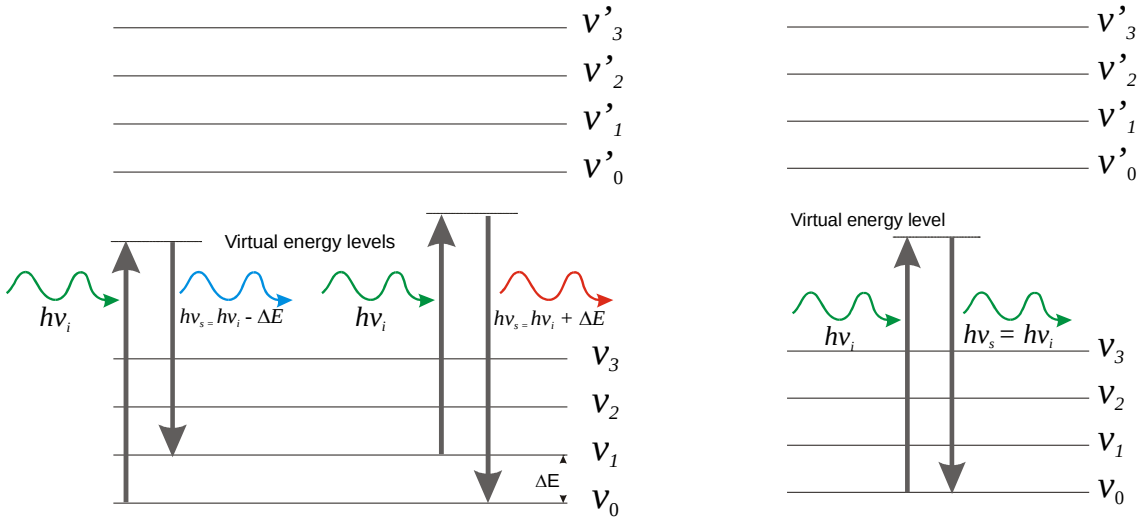


Fig. 2. The electronic transitions accompanying Raman scattering (left), Rayleigh scattering (right)

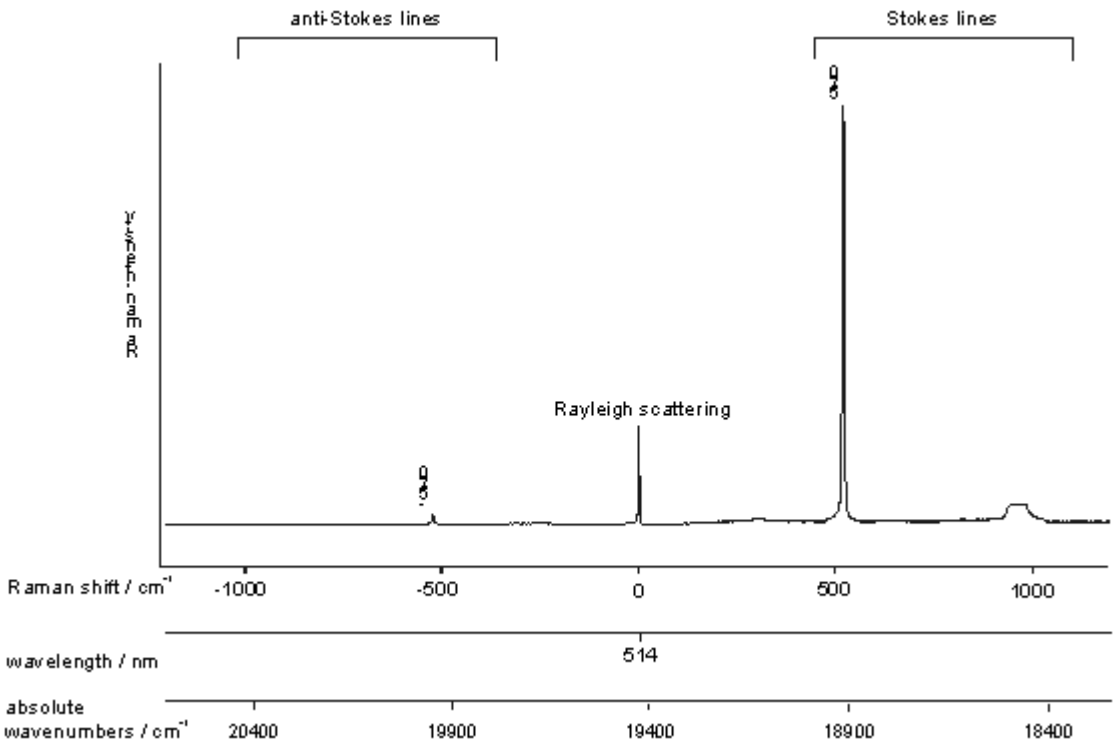


Fig. 3. Raman spectrum of silicon (514 nm excitation) showing the Rayleigh scattering at the laser wavelength and the Stokes and anti-Stokes line of the Raman scattering

What does a Raman spectrum look like?

Figure 4 shows the Raman spectra of two carbon based species, diamond and polystyrene. In Raman spectroscopy we are interested in how much the scattered light differs from the incident light, so the spectrum is normally plotted against the difference between the two - the Raman shift.

Diamond has one main Raman band only because the tetrahedral lattice is symmetrical and all the carbon atoms and connecting bonds are equivalent.

Polystyrene has different functional groups consisting of differing atoms and bond strengths. Each Raman band represents either a discrete function group (e.g. C-H from benzyl group at ~3200 cm^{-1}) or a combining of small groups into a larger group (e.g. C_6H_5R breathing mode from benzyl group at ~1000 cm^{-1}).

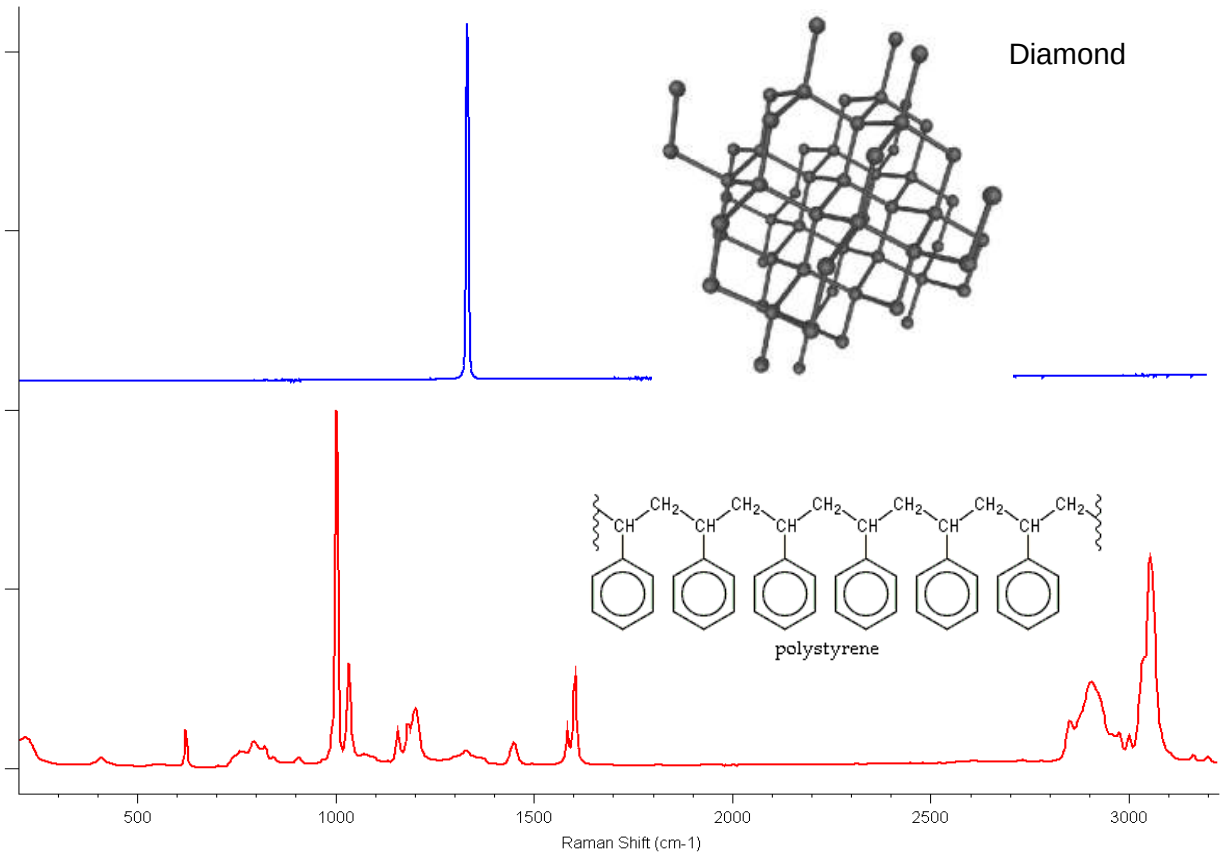


Fig. 4. Raman spectra of diamond and polystyrene

The bottom axis of the graph represents the energy of the Raman shift (measured in cm^{-1}) and may be plotted right-to-left or *vice versa*. A value of 0 cm^{-1} would indicate that no energy has been exchanged with the sample and the incident light is scattered with no change in wavenumber. Carbon-hydrogen bonds give rise to Raman bands around 3000 cm^{-1}, due to the small mass of hydrogen and resulting high frequency vibrations. Peaks at lower wavenumber relate to lower energy vibrations such as those of bonds to carbon or oxygen.

What information can you get from Raman spectroscopy?

Raman bands can be analysed to obtain chemical and structural information for the material identification, investigation of material properties and spatial analysis. The table below illustrates the variety of results that can be obtained from point and mapping measurements.

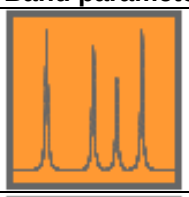
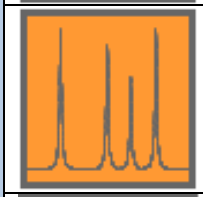
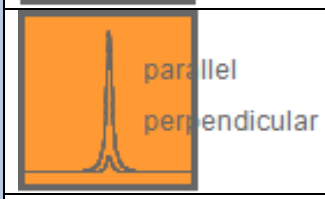
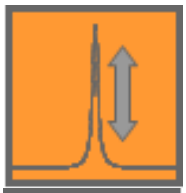
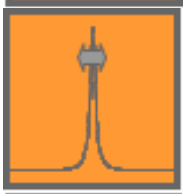
	Band parameter		Information
Univariate		Characteristic Raman frequencies	Identification (material composition)
		Compare characteristic Raman frequencies	Differentiation
		Intensity variation with changing polarisation	Crystallographic orientation
		Variation in absolute / relative intensity	Absolute / relative concentration
		Variation in Raman band width	Crystallinity; Temperature
		Variation in Raman band position	Stress state
Multiple spectra	Any of the above parameters applied to multiple spectra: • Univariate – based on raw data or curve fitting. • Multivariate – algorithm based *e.g.* DCLS, PCA, MCR-ALS or EmptyModelling™.		Any of the above information in conjunction with different dimensions such as time, temperature, distance, area, and volume, *e.g.* • Thickness (Intensity with depth – 1D) • Domain size and distribution (Intensity with area – 2D / 3D)

Table 1. Information obtainable from analysis of different Raman band parameters.

Raman spectroscopy obtains such information by probing the vibrational states of materials. Renishaw's inVia can also be used for photoluminescence (PL) measurements, which is a competing effect to Raman. PL is typically much stronger in intensity and is a function of the electronic states of the material. The PL effect can sometimes provide an unwanted broad background that can mask the Raman bands. However, PL measurements can also provide useful complementary information on material properties such as conjugation, structural vacancy, and atomic substitutions.

What does a micro-Raman instrument usually consist of?

It usually consists of:

- A monochromatic light source (normally a laser)
- A means of shining the light on the sample and collecting the scattered light (often this is a microscope)
- A means of filtering out all the light except for the tiny fraction that has been Raman scattered (often holographic 'notch' or dielectric 'edge' filters)
- A device (such as a diffraction grating) for splitting the Raman scattered light into component wavelengths, i.e. a spectrum.
- A light-sensitive device for detecting this light (normally a CCD camera)
- A computer to control the instrument and the motors and analyse and store the data

Figure 5 shows the layout of Renishaw's inVia Reflex Raman microscope, with all the key components highlighted.

Fig. 5. Renishaw's inVia Reflex Raman microscope